Econ 812 Recitation 9 Notes

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1 Hints for the next PS

1.1 Question 1

Lucas human capital model is from the example below

1.2 Question 2

For the golden rule level of capital choose k^* to maximize consumption.

1.3 Question 3

See below on two sector model

1.4 Question 4

This is a two sector model, you get the 2 EE with $x = k/g$ and can solve for x.

1.5 Question 5

Remember that Y is total output while $y := Y/L$ is output per capita, so $y_{us}/y_I = 10$.

2 Steps for Solving Endogenous Growth Problems

The overall goal here is to get the growth rates in terms of constants, and that the growth rates of output, consumption, capital should be the same, and wages should be constant (no growth). Also, the constraints on the parameters should be that $\gamma_c > 1$. So whenever you get the growth rate in terms of constants, plug them back into expression for growth rate and say it needs to be greater than 1.

1. Set up HH lagrangian

2. Take FOCs

3. Solve for EE (γ_c)

4. [1]Use other equations and definitions from problem to solve for growth rate. Then set the EE equal to the growth rate found from definitions in problem to solve for constant ratio of endogenous variables, or, if not applicable then you should get the growth rate in terms of parameters.

5. Use firm's production function to find a relationship between γ_y and γ_k

6. Say that by the BC and TVC $\gamma_c = \gamma_y = \gamma_k$.

3 Endogenous Growth Model August 23

There is a continuum of households with representative household preferences over consumption given in equation 1.

$$\sum_{t=0}^{\infty} \beta^t \ln C_t \tag{1}$$

Where C_t is consumption. The household has 1 unit of time that it can supply to either production or the accumulation of human capital according to the law of motion:

$$H_{t+1} = GH_t(1 - N_t)$$

And physcial capital evolves according to:

$$K_{t+1} = I_t$$

with $K_0 > 0$ given. The household's budget constraint is

$$C_t + I_t = r_t K_t + w_t H_t N_t + \pi_t$$

where π_t is any profit earned by the firms - taken as given by the households. There is a unit continuum of firms. The representative firm has a production function given by

$$y_t = \Gamma k_t^{1/2} z_t^{1/2}$$

with $\Gamma > 0$. Market clearing in the goods market requires $C_t + I_t = Y_t$.

[1]Note here that solving these problems sometimes is context specific. If you see something like $A_t = Y_{t-1}^{1/2}$ pretty good chance you should write out the definition of γ_A, and this should help you solve for the ratio of endogenous variables (think z/k or A/k).

3.1 Part A

Define a symmetric competitive equilibrium for this economy. [5 pts]

A symmetric competitive equilibrium is a set of prices $P = \{w_t, r_t\}$, a plan for the firm $F = \{k_t, z_t, \pi_t\}$ a plan for the households $H = \{K_t, I_t, C_t, N_t, H_t\}$ subject to given prices and an initial amount of capital, H solves the household's problem, F solves the firm's problem, goods markets clear $Y_t = C_t + I_t$, capital markets clear $k_t = K_t$ and labor markets clear $z_t = N_t H_t$

3.2 Part B

Under what conditions on model parameters do we have a balanced growth path with positive growth. Under these conditions, solve for the balanced growth path. At what rates do consumption, wages, and output grow along the balanced growth path? [20 pts]

Set up the household's problem:

$$\mathscr{L} = \sum_{t=0}^{\infty} \beta^t [\log C_t + \lambda_t [-C_t - K_{t+1} + r_t K_t + w_t H_t N_t] + \mu_t [G H_t (1 - N_t) - H_{t+1}]] \tag{2}$$

Taking FOC's

$$C_t : 1/C_t = \lambda_t$$

$$K_{t+1} : \lambda_t = \beta \lambda_{t+1} [r_{t+1}]$$

$$N_t : \lambda_t w_t = \mu_t G$$

$$H_{t+1} : \beta \lambda_{t+1} w_{t+1} N_{t+1} + \beta \mu_{t+1} G (1 - N_{t+1}) = \mu_t$$

Plus the BC, law of motion for human capital, and TVC.

Find the EE to be

$$1/C_t = \beta 1/C_{t+1}(r_{t+1}) \implies \gamma_c = \beta r_{t+1}$$

and

$$\beta 1/C_{t+1} w_{t+1} N_{t+1} + \beta 1/C_{t+1} w_{t+1} (1 - N_{t+1}) = 1/G C_t w_t$$

$$\gamma_c = G\beta > 1$$

From the firm's problem we have $r_{t+1} = 1/2\Gamma k_{t+1}^{-1/2} z_{t+1}^{1/2}$ Substituting to find

$$\gamma_c = \beta 1/2\Gamma (z/k)^{1/2} > 1$$

As before, this implies that the ratio of z/k is constant on a balanced growth path. This implies that $\gamma_z = \gamma_k$

$$G\beta = \beta 1/2\Gamma(z/k)^{1/2} \implies (z/k)^{1/2} = \frac{2G}{\Gamma}$$

From the firm's production function we have

$$\gamma_y = \gamma_k^{1/2}\gamma_z^{1/2} \implies \gamma_y = \gamma_k = \gamma_z$$

And by the RC and TVC we say that $\gamma_c = \gamma_y$. By definition we also have that $\gamma_z = \gamma_H\gamma_N$, and from the law of motion of human capital we have: $\gamma_H = G(1 - N_t)$. This implies that N is constant, you can think of this as investing a fixed fraction of your time into human capital accumulation (education) in order to grow your human capital at a constant rate. Additionally, from the definition of $w_t = 1/2\Gamma(k/z)^{1/2}$ we have that $\gamma_w = 1$.

3.3 Part C

How does an increase in β affect the growth rate and level of output in this economy? Explain. How does an increase in G affect the growth rate and level of output in this economy? Explain. [16 pts]

An increase in β or G would increase γ_c and increase the level of output in the economy. This makes senese since valuing future consumption more increases the incentive to invest today (increasing output), and increasing the returns to education makes the labor force more productive.

3.4 Part D

Is the competitive equilibrium Pareto optimal? Explain. [9 points]

It would be Pareto optimal if the SPP leads to the same outcome. In that case

$$\mathscr{L} = \sum_{t=0}^{\infty} \beta^t [\log C_t + \lambda_t[-C_t - K_{t+1} + \Gamma K_t^{1/2} H_t^{1/2} N_t^{1/2}] + \mu_t[GH_t(1-N_t) - H_{t+1}]] \tag{3}$$

Taking FOC's:

$$C_t : 1/C_t = \lambda_t$$

$$K_{t+1} : \lambda_t = \beta\lambda_{t+1}[\Gamma 1/2 K_{t+1}^{-1/2}(H_t N_t)^{1/2}]$$

$$N_t : \lambda_t 1/2\Gamma K_t^{1/2} H_t^{1/2} N_t^{-1/2} = \mu_t G H_t$$

$$H_{t+1} : \beta\lambda_{t+1}\Gamma 1/2 K_{t+1}^{1/2} H_{t+1}^{-1/2} N_{t+1}^{1/2} + \beta\mu_{t+1}G(1 - N_{t+1}) = \mu_t$$

Plus the BC, law of motion for human capital, and TVC.

Combine to get

$$\gamma_c = \beta 1/2\Gamma(HN/K)^{1/2}$$

and

$$\beta 1/C_{t+1}\Gamma 1/2K_{t+1}^{1/2}H_{t+1}^{-1/2}N_{t+1}^{1/2}+\beta 1/C_{t+1}\Gamma 1/2K_{t+1}^{1/2}H_{t+1}^{-1/2}N_{t+1}^{-1/2}(1-N_{t+1}) = 1/C_t 1/2\Gamma/GK_t^{1/2}H_t^{-1/2}N_t^{-1/2}$$

$$\beta 1/C_{t+1}(K_{t+1}/Z_{t+1})^{1/2}N_{t+1}+\beta 1/C_{t+1}(K_{t+1}/Z_{t+1})^{1/2}(1-N_{t+1}) = 1/C_t(K_t/Z_t)^{1/2}1/G$$

And the ratio $(K_{t+1}/Z_{t+1})^{1/2} = (K_t/Z_t)^{1/2}$ so we reduce to:

$$\gamma_c = \beta G$$

These are the same as before so yes it is Pareto optimal.

4 Two sector model

There are two goods, consumption and investment goods. You can only consume a taxed portion of what is produced by the production function $Ak_{1,t}^{\alpha}$. The investment goods evolve according to law of motion and $k_{t+1} = bk_{2,t}$ and the two types of capital is also taxed and sum up as $k_t = k_{1,t}+k_{2,t}$. Utility is maxed by:

$$\sum_{t=0}^{\infty} \beta^t \log(c_t)$$

Set up the lagrangian to be

$$\mathscr{L} = \sum_{t=0}^{\infty} \beta^t[\log(c_t)+\lambda_t[(1-\tau)Ak_{1,t}^{\alpha}-c_t]+\mu_t[bk_{2,t}-k_{t+1}]+\sigma_t[(1-\tau)k_t-k_{1,t}-k_{2,t}]]$$

Take FOCs

$$c_t : 1/c_t = \lambda_t$$

$$k_{1,t} : \lambda_t(1-\tau)A\alpha k_{1,t}^{\alpha-1} = \sigma_t$$

$$k_{2,t} : \mu_t b = \sigma_t$$

$$k_{t+1} : \mu_t = \beta(1-\tau)\sigma_{t+1}$$

Combine to get

$$\frac{1}{c_t}(1-\tau)A\alpha k_{1,t}^{\alpha-1} = b\beta(1-\tau)\frac{1}{c_{t+1}}(1-\tau)A\alpha k_{1,t+1}^{\alpha-1}$$

Reduce to

$$\gamma_c = b\beta(1-\tau)\gamma_{k_1}^{\alpha-1}$$

From the production function we have

$$Y_t = Ak_{1,t}^{\alpha}$$

So dividing by the production function in $t-1$ we have

$$\gamma_y = \gamma_{k_1}^{\alpha}$$

From the investment goods law of motion we have $\gamma_k = \gamma_{k_2}$
Then from the B.C. we have $c_t = (1-\tau)Ak_{1,t}^{\alpha}$ implies

$$\gamma_c = \gamma_{k_1}^{\alpha} = \gamma_y$$

So from the E.E we have that

$$\gamma_{k_1}^{\alpha} = b\beta(1-\tau)\gamma_{k_1}^{\alpha-1}$$

Solve to find

$$\gamma_{k_1} = b\beta(1-\tau)$$